WORKSTATION Study

Time Analysis

1. Statement

You are responsible for carrying out the following workstation study in order to propose improvements to your management.

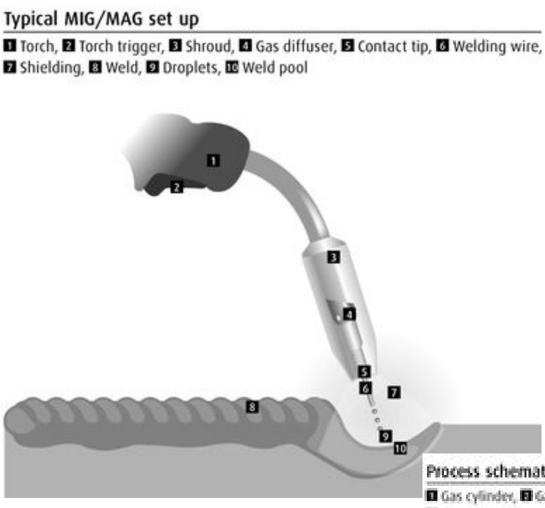
- Workshop: Welding
- Operator concerned: Martin
- Manufacturing process plan: Assembled frame
- Annual quantity: 4,200 parts
- Release Qty: 180 parts
- Hourly rate: 43€
- Current welding operation time T = 663 dmh (= 238.68sec.)
- Time of the different operations will be given according to your needs
- Welding operation according to the MAG process (described below)

2. Goal

Your objective is to respect the cost and time required:

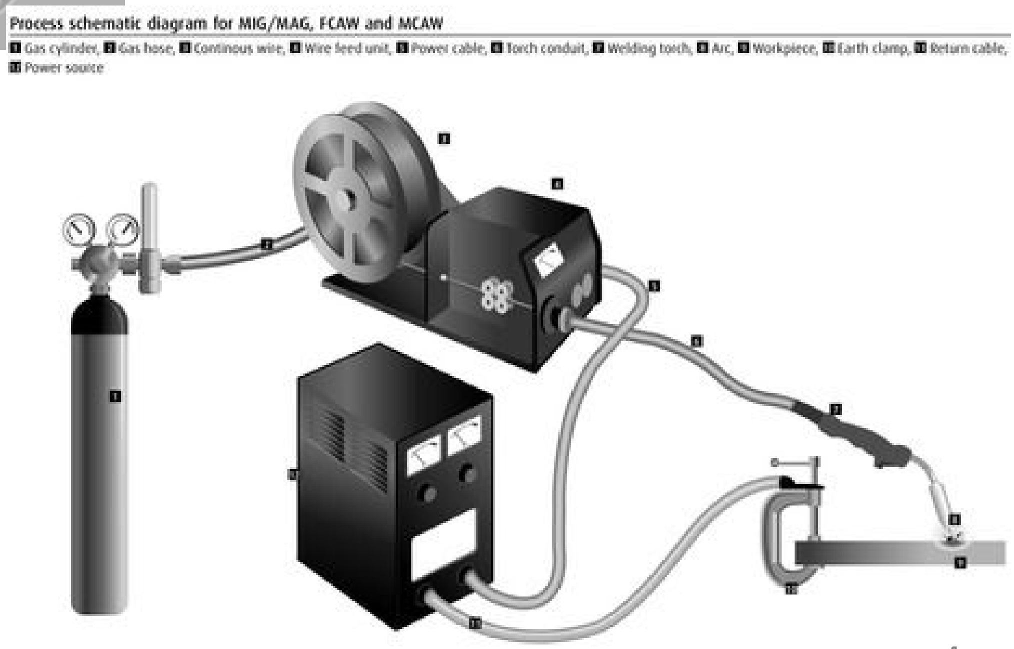
- Takt Time: 555 dmh (=200sec.)
- Cost not to be exceeded: 4000€
- Improve the welding workstation by changing the welding helmet to free the hand for welding

Gas metal arc welding (GMAW) description:



Metal Inert Gas (MIG) / Metal Active Gas (MAG) welding refers to a group of arc welding processes that use the heat generated by a DC electric arc to fuse the metal in the joint area. A continuous electrode (the wire) is fed by powered feed rolls (wire feeder) into the weld pool.

An electric arc is created between the tip of the wire and the weld pool. The wire is progressively melted at the same speed at which it is being fed and forms part of the weld pool. Both the arc and the weld pool are protected against atmospheric contamination by a shield of inert (non-reactive) gas.



The MIG/MAG process is suited to a variety of applications provided the shielding gas, electrode (wire) size and welding parameters have been correctly set. Welding parameters include the voltage, travel speed, arc (stick-out) length and wire feed rate. The arc voltage and wire feed rate will determine the filler metal transfer method.

(Source: https://www.linde-gas.com)

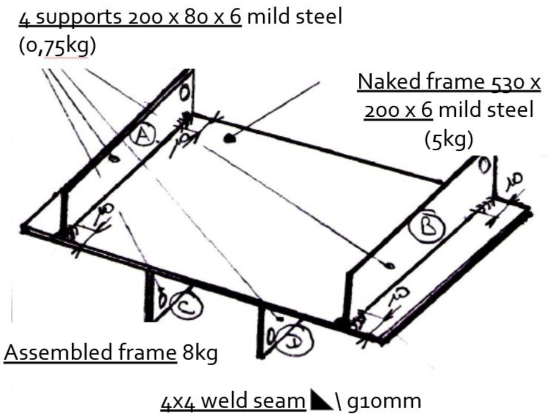
3. Data available

<u>You will also find:</u>

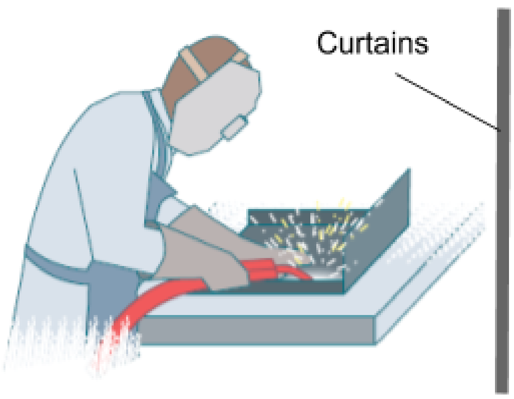
- A photo of the old generation welding generator and welding mask (must be held by one hand) that are currently in use

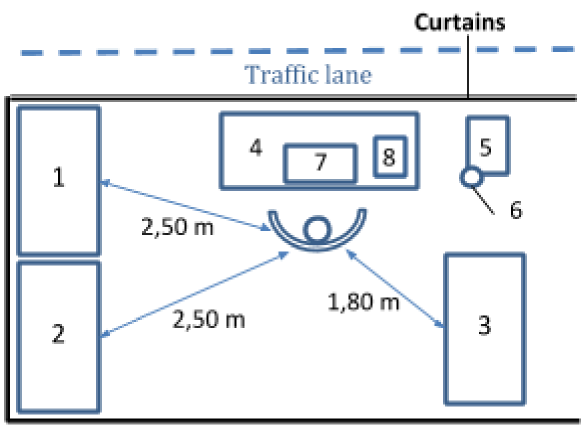
- The new generation welding mask (helmet) to be used

- A sketch of the assembled frame (cf **Diagnostic Sheet No. 1**)



- A sketch of the workstation N°6 (cf **Diagnostic Sheet No. 1**)





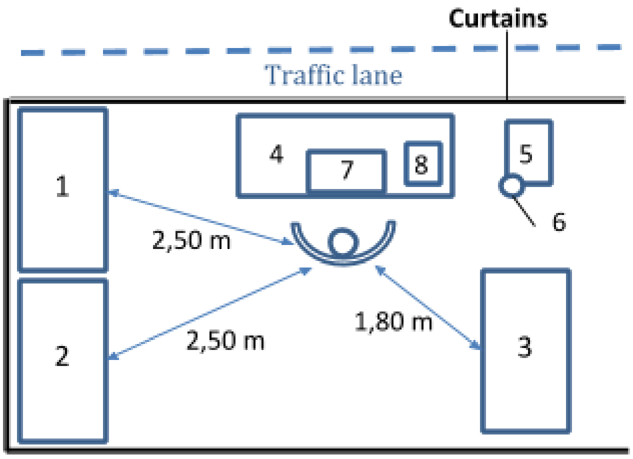
Diagnostic Sheet No. 1: Workstation information

DIAGNOSTIC - STUDY - PROJECT PHASE 1

Study N°	1	**Workshop**	Welding	**Workstation N°**	06
Date	11/08/2020	**Operator**	Martin	**Qualification**	ISO 9606
Name	Dupont	**Process plan**	Assembled frames		
Visa		**Operation**	MAG welding		

PRODUCTION		INITIAL STATE	FINAL STATE
Annual	Serial	6 mm thick sheet metal in oxycutting	Welded assembly
4 200	180		

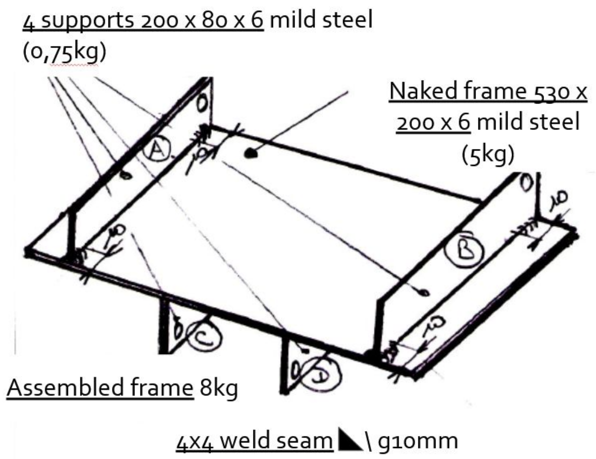
PRE-STUDY WORKSTATION SKETCH



EQUIPMENTS

NUMBER	DESIGNATION
1	Naked frame container
2	ABCD support container
3	Container assembled frame
4	Worktable
5	MAG welding station
6	Welding torch holder
7	Welding tools on table (plate + locating + clamp)
8	Welding mask

PRODUCT SKETCH



The 22 rules of economy of movements:

Simultaneity of movements
1. Both hands must start and finish their movements at the same time.
2. Both hands must not remain inactive at the same time, except when resting.
3. Arm movements must be symmetrical and simultaneous.

Minimum energy expenditure
4. The movements required for the work must use the smallest possible muscle masses.
5. Continuous movements are preferable to "Zig Zag" movements or broken line movements with acute angles.

The living force
6. Living force should be used whenever possible to assist the operator's movements. It shall be reduced to a minimum if movement is controlled.
7. Ballistic movements are faster, easier and more accurate than constrained or controlled movements.

Pace
8. The acquisition of a pace is essential for the easy and automatic execution of a work.

Order of the equipped work area
9. There must be a defined place for all materials or components.
10. Tools, materials and checkers should be placed as close as possible and as far in front of the operator as possible.
11. Materials, components and tools must be arranged to allow the best possible sequence of movements.

Use of gravity
12. Gravity feed boxes and containers must supply the performer(s) in close proximity to their workplace.
13. Gravity must be used for evacuation: chutes, conveyors, inclined rollers, etc.

Comfort and lighting of the work surface
14. Each operator must be provided with the best conditions for lighting his work.
15. The height of the work surface and the seat must, as far as possible, allow work to be carried out either standing or sitting down.
16. A seat allowing good posture must be provided for each operator.

Freedom of hands
17. Hands must be relieved of all work that can be done more advantageously by assembly.

Combining / positioning
18. Tools should be combined whenever possible.
19. Tools and equipment should be positioned whenever possible.

Finger loading
20. When each finger performs a separate movement, the load of each finger must be distributed according to the capabilities of each.

Control elements
21. Handholds shall allow the largest possible contact area with the surface of the vehicle.
22. Levers, capstans and steering wheels must allow them to be manoeuvred with the least change in posture and with the greatest possible mechanical efficiency.

**The elements in green are the rules that can be applied in our case of use.*

4. Exercises

To carry out this study, you will use the "Standardized Work Combination Table (SWCT)" document to diagnose the workstation No.6 in order to suggest improvements:

1. Establish the current state of the welding process by filling the SWCT. ***Pay attention to the conversion of the units dmh <> sec !*** (complete **Diagnostic Sheet No. 2**)
2. Analyse the SWCT and suggest improvements that respect the initial goal of the takt time, the cost and the 22 rules of economy of movements (complete **Diagnostic Sheet No. 3**)
3. With the estimated time savings, re-establish a SWCT, to validate the proposed solution (complete **Diagnostic Sheet No. 4**)

Reminder:

	dmh	h	mn	sec
seconde	100/36	1/3600	1/60	1

Sequence of operations*:

OPERATION 10 **WELDING 1st SIDE**

- Op. 11 - Go take the frame in container and place on table: 42dmh
- Op. 12 - Installing the frame in the tooling: 27dmh
- Op. 13 - Go take ABCD support in the container: 30dmh
- Op. 14 - Positioning / clamping supports AB: 35dmh
- Op. 15 - Grab mask + torch: 22dmh
- Op. 16 - Solder 8 cords 10 mm: 133dmh
- Op. 17 - Put on the mask + torch: 13dmh
- Op. 18 - Debriding/brushing: 44dmh

OPERATION 20 **WELDING 2nd SIDE**

- Op. 21 - Turning / positioning the frame in the tooling: 22dmh
- Op. 22 - Positioning/clamping supports CD: 35dmh
- Op. 23 - Grab mask + torch: 22dmh
- Op. 24 - Solder 8 cords 10 mm: 133dmh
- Op. 25 - Put on the mask + torch: 13dmh
- Op. 26 - Debriding/brushing: 44dmh

OPERATION 30 **EVACUATE ASSEMBLED FRAME**

- Op. 31 - Grab the frame: 8dmh
- Op. 32 - Evacuate the frame into the container: 40dmh

****Based on a minimum of 5 observations. The results of the observed times are the average of the samples***

Diagnostic Sheet No. 2: Standardized Work Combination Table CURRENT STATE

Standardized Work Combination Table

OPERATOR NAME:

DEPARTMENT:

STUDY N°:

WORKSHOP:

WORKSTATION:

PROCESS PLAN:

PRODUCT DESIGNATION:

OPERATION NAME:

INVESTIGATOR NAME:

PROCESS ANALYSIS

Op. N° : to to

Op. N° :

Takt Time:

Cycle Time:

Qty / shift:

Available time:

VISA:

DATE:

MANUAL

AUTOMATIC

WALKING

DEVIATION

WAITING

SEQ. ORDER	WORK SEQUENCE	Waiting	SEQUENCE DURATION manual	automatic	walking	WORK SEQUENCE DIAGRAM 20' 40' 60' 80' 100' 120' 140' 160' 180' 200' 220' 240' 260' 280' 300'
		SEC	SEC	SEC	SEC	
	Cumulative Time					Operator Cycle Time = Deviation = Nb parts / Cycle = Operator waiting time = Cycle Time =

Diagnostic Sheet No. 3: Assessment

DIAGNOSTIC - STUDY - PROJECT PHASE 2	
SKETCH OF THE POSITION/EQUIPMENT TOOLS BEFORE STUDY	SKETCH OF THE POSITION/EQUIPMENT TOOLS AFTER STUDY

ESTIMATED BALANCE SHEET					
	BENEFIT			EXPENSES	

RETURN ON INVESTISSEMENT

KEY CRITERIA

STUDY N°		NAME		VISA		PAGE /

Diagnostic Sheet No. 4: Standardized Work Combination Table FUTURE STATE

OPERATOR NAME:
DEPARTMENT:
STUDY N°:

Standardized Work Combination Table

WORKSHOP:
WORKSTATION:
PROCESS PLAN:
PRODUCT DESIGNATION:
OPERATION NAME:

INVESTIGATOR NAME:
PROCESS ANALYSIS
Op. N°: to
to to
Op. N°:

Takt Time:
Cycle Time:
Qty / shift:
Available time:

VISA:
DATE:

MANUAL
AUTOMATIC
WALKING
DEVIATION
WAITING

SEQ. ORDER
WORK SEQUENCE
Waiting SEC
SEQUENCE DURATION
manual SEC
automatic SEC
walking SEC

WORK SEQUENCE DIAGRAM
20' 40' 60' 80' 100' 120' 140' 160' 180' 200' 220' 240' 260' 280' 300'

Cumulative Time

Operator Cycle Time =
Deviation =
Nb parts / Cycle =
Operator waiting time =
Cycle Time =